

Vitis Release Notes And Installation Guide

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Vitis Release Notes

What's New

For information about what's new in this version of the AMD Vitis™ unified software development platform, see the [Vitis What's New Page](#).

Embedded GNU Toolchain

The following GNU toolchain components are installed with the Vitis unified software platform.

- **binutils:** 2.42
 - **gcc:** 13.2
 - **gdb:** 14.2
 - **glibc:** 2.39
 - **newlib/libgloss:** 4.4.0
 - **linux-libc-headers:** 6.6
-

Supported Platforms [Vitis Accelerated Software Development]

This section details the supported platforms for Vitis Accelerated Software Development.

Data Center Accelerator Cards

Access the latest Vitis target platforms for AMD Alveo™ Data Center accelerator cards at www.xilinx.com/products/boards-and-kits/alveo.html.

Refer to the *Alveo Data Center Accelerator Card Platforms User Guide* ([UG1120](#)) for specifications of each accelerator card and available target platforms. The *Getting Started* section for each accelerator card has information for deploying your applications on that card.

Refer to *Installing Xilinx Runtime and Platforms in the Vitis High-Level Synthesis User Guide* ([UG1399](#)) for more information on setting up XRT and platforms.

Vitis Embedded Platforms

The Vitis installer includes a selection of pre-built embedded base platforms that serve as a starting point for development. The following platforms are provided:

- `xilinx_vek280_base_202420_1`
- `xilinx_vck190_base_202420_1`
- `xilinx_vck190_base_dfx_202420_1`
- `xilinx_vmk180_base_202420_1`
- `xilinx_zcu102_base_202410_1`
- `xilinx_zcu102_base_dfx_202420_1`
- `xilinx_zcu104_base_202420_1`

The source code for Vitis base platforms is available on GitHub: https://github.com/Xilinx/Vitis_Embedded_Platform_Source

In addition, users have the flexibility to create custom Vitis embedded platforms to suit specific project requirements.

Versal Platforms for AI Engine Development

The VCK190 and VEK280 platforms are available for use with the Vitis application acceleration development flow, as described in the *AI Engine Tools and Flows User Guide* ([UG1076](#)). The platform enables development of designs that include:

- AI Engine graphs and kernels.
- Programmable logic kernels.
- Host application targeting the Linux or a bare metal OS running on the Arm processor in the Versal device.

Common Image for Vitis Embedded Platforms

Common images for Vitis embedded base platforms can be downloaded from the [Embedded Platforms download page](#). These images include pre-built components such as the Linux kernel, root filesystem (rootfs), sysroot, and boot files for each supported device family. They provide a streamlined setup for Vitis base platform software, allowing users to concentrate on application development rather than system configuration. These software components can be customized as needed to meet specific development needs.

Changed Behavior

The following table specifies differences between this release and prior releases that impact behavior or flow when migrating.

Table 1: Changed Behavior Summary for Vitis Software Development Platform

Scope	Details
Vitis Unified IDE	Metadata files for the IDE have been reorganized. A new file, <code>_ide/.wsdata/version.ini</code> , has been introduced to label the workspace version. When an older workspace or a workspace without <code>version.ini</code> is opened, the IDE will prompt the user to update the workspace. The reorganization targets to help users to add workspace to source control tool easier by keeping files can be regenerated in the <code>.wsdata</code> folder.
Vitis Unified IDE	The Python journal file <code>builder.py</code> , previously located in the logs directory, has been renamed to <code>workspace_journal.py</code> and moved to the <code>_ide</code> directory. It can now be accessed from the menu under Vitis → Workspace Journal.
Vitis Unified IDE	The IDE now enables users to manage application component source code compilation settings in <code>UserConfig.cmake</code> . Previously, the IDE automatically set up the build list.
Vitis Unified IDE	The process name of Vitis IDE is changed from <code>vitisng-ide</code> to <code>vitis-ide</code> .
Vitis Unified IDE	The user managed flow is extended from the Explorer view to the Vitis Explorer view. Vitis Explorer can handle both the GUI managed flow and the user managed flow and provide consistent flow window support.

Table 2: Changed Behavior Summary for Vitis Accelerated Software Development

Scope	Details
Vitis Base Platforms	The VCK190 Base Platform now initiates designs using the Vivado Customizable Example Design (CED) Vitis Extensible Platform rather than building from scratch. This update aligns the Vitis platform configuration with both CED and Vitis Base Platforms. The same approach will be applied to other base platforms in future releases.

Deprecated Features

The following features are deprecated as of this release.

- **Classic Vitis IDE:** The Classic Vitis IDE is deprecated and will be removed in the 2025.1 release.
- **XSCT (Xilinx Software Command-line Tool):** XSCT is deprecated and will be removed in future releases. It is recommended to use the Vitis Python API for project management. For more information, refer to [Python API](#). For Tcl based commands for debugging, XSDB provides the debugging support. For more details, refer to *Software Debugger Reference Guide* ([UG1725](#)). For project management features previously available in XSCT, refer to earlier versions of Vitis *Unified Software Platform Documentation: Embedded Software Development* ([UG1400](#)).
- **Software Emulation for Acceleration Development Flow:** This feature is deprecated. For more information, refer to the support article: [000036790 - Deprecation of Software Emulation in Vitis Software Platform](#).
- **OpenCL as Vitis PL Kernel Source:** Using OpenCL as the PL kernel source in Vitis is deprecated. This feature will be removed in the 2025.1 release.

Known Issues

Known issues for the Vitis software platform are available on the [Vitis 2024 Known Issues](#) web page.

Installing the Vitis Platform

Installation Requirements

The AMD Vitis™ unified software platform consists of an integrated design environment (IDE) for interactive project development, and command-line tools for scripted or manual application development.

To install and run on a PC, your system must meet the following minimum requirements.

Table 3: Application Acceleration Development Flow Minimum System Requirements

Component	Requirement	
	Development (Build Machine OS)	Deployment (Host OS) Enabled via XRT
Operating system	Linux, 64-bit: - Red Hat Enterprise Linux Workstation/Server 7/CentOS: 7.9 - Red Hat Enterprise Linux Workstation/Server: 8.8, 8.9, 8.10 - Red Hat Enterprise Linux Workstation/Server: 9.2, 9.3, 9.4 - AlmaLinux: 8.7, 8.10, 9.1, 9.4 - Ubuntu 20.04.5 LTS, 20.04.6 LTS - Ubuntu 22.04.1 LTS, 22.04.2 LTS, 22.04.3 LTS, 22.04.4 LTS - Ubuntu 24.04 LTS - Amazon Linux 2 AL2 LTS Windows, 64-bit for Embedded Software and HLS Development: - Windows 10: 22H2 - Windows 11: 23H2	For on-premise acceleration (Alveo Data Center accelerator cards): - Red Hat Enterprise Linux Workstation/Server 7/CentOS: 7.9 - Red Hat Enterprise Linux Workstation/Server: 8.8, 8.9, 8.10 - Red Hat Enterprise Linux Workstation/Server: 9.2, 9.3, 9.4 - AlmaLinux: 8.7, 8.10, 9.1, 9.4 - Ubuntu 20.04.5 LTS, 20.04.6 LTS - Ubuntu 22.04.1 LTS, 22.04.2 LTS, 22.04.3 LTS, 22.04.4 LTS - Ubuntu 24.04 LTS - Amazon Linux 2 AL2 LTS - Linux in VM on Windows Server 2019 For edge acceleration (embedded platforms): - PetaLinux 2024.2
System memory	For Alveo cards: 64 GB (80 GB is recommended) For embedded: 32 GB	
Internet connection	Required for downloading drivers and utilities.	
Hard disk space	150 GB	

Note: The Vitis application acceleration development flow is not supported for the Windows OS. However, the Vitis IDE runs on Windows OS to support both the embedded software flow, and high-level synthesis (HLS) development flow.

Note: Amazon Linux 2 AL2 LTS does not support AI Engine development.

Table 4: Embedded Software Development Flow Minimum System Requirements

Component	Requirement
	Development (Build Machine OS)
Operating system	Linux, 64-bit: - Red Hat Enterprise Linux Workstation/Server 7/CentOS: 7.9 - Red Hat Enterprise Linux Workstation/Server: 8.8, 8.9, 8.10 - Red Hat Enterprise Linux Workstation/Server: 9.2, 9.3, 9.4 - AlmaLinux: 8.7, 8.10, 9.1, 9.4 - Ubuntu 20.04.5 LTS, 20.04.6 LTS - Ubuntu 22.04.1 LTS, 22.04.2 LTS, 22.04.3 LTS, 22.04.4 LTS - Ubuntu 24.04 LTS - Amazon Linux 2 AL2 LTS - SUSE Enterprise Linux 15.4 Windows, 64-bit: - Windows 10: 22H2 - Windows 11: 23H2
System memory	32 GB (64 GB is recommended)
Internet connection	Required for downloading drivers and utilities.
Hard disk space	- Full Vitis installation: 200GB - Vitis Embedded Development: 25GB

Operating System End-of-life Announcement

The following operating systems will not be supported in the next release of the Vitis Software Platform. Recommend upgrade to latest operating system as early as possible.

- Red Hat Enterprise Linux Workstation/Server 7/CentOS: 7.9
- Red Hat Enterprise Linux Workstation/Server: 8.8, 8.9
- AlmaLinux: 8.7, 9.1
- Ubuntu 20.04.5 LTS, 20.04.6 LTS
- Ubuntu 22.04.1 LTS

Install Required RHEL/CentOS/AlmaLinux Packages

Before installing the Vitis software platform on CentOS or RedHat, you must install the Extra Packages for Enterprise Linux (EPEL), and ensure you have the proper kernel-headers and kernel-devel packages installed. The initial setup commands depend on the operating system and version you are using. For more information, see <https://fedoraproject.org/wiki/EPEL>.

1. Install EPEL.

- **On RedHat/CentOS/AlmaLinux:**

For RHEL/CentOS 7, open a terminal window and enter the following command:

```
$ sudo yum install https://dl.fedoraproject.org/pub/epel/epel-release-latest-7.noarch.rpm
```

On RHEL 7, enable the `-optional`, `-extras`, and `-ha` repositories because EPEL packages might depend on packages from these repositories:

```
$ sudo subscription-manager repos --enable "rhel-*-optional-rpms" --enable "rhel-*-extras-rpms" --enable "rhel-ha-for-rhel-*-server-rpms"
```

- **On CentOS:**

Alternatively, on CentOS open a terminal window, and enter the following command:

```
sudo yum install epel-release
```

2. To install kernel headers and kernel development packages, run the following commands:

```
$ sudo yum install kernel-headers-$(uname -r)`  
$ sudo yum install kernel-devel-$(uname -r)`
```

Note: Ensure that `uname` is surrounded by backticks (```) and *not* single quotes (`'`).

3. To install additional packages required for emulation, run the following commands.

```
sudo yum install gcc  
sudo yum install gcc-c++
```

4. Cold reboot your system.

Installing Required Ubuntu Packages

Install the following packages *before* you install the Vitis software platform.

1. Install the packages with the following command.

```
sudo add-apt-repository ppa:xorg-edgers/ppa  
sudo apt-get update  
sudo apt-get install libgl1-mesa-glx  
sudo apt-get install libgl1-mesa-dri  
sudo apt-get install libgl1-mesa-dev
```

```
sudo add-apt-repository --remove ppa:xorg-edgers/ppa
sudo apt install net-tools
sudo apt-get install -y unzip
sudo apt install gcc
sudo apt install g++
```

2. Cold reboot your system.

Installing Additional Packages for Versal AI Engine

On CentOS/RHEL/Ubuntu machines, when targeting the AI Engine tools, you must install additional packages.



TIP: The script will change the user environment as root, so you should review it before running.

1. Navigate to `<install_directory>/xinstall/scripts`, and then run `installAIeDepLibs.sh`.
2. Set up the symbolic links:

```
cd /usr/lib64
sudo ln -s libexpat.so.1.6.0 libexpat.so.0
sudo ln -s libexpat.so.1.6.0 libexpat.so
```

Vitis Software Platform Installation

Installing the Vitis Software Platform

1. Go to the [AMD Adaptive Computing Downloads Website](#).
2. Download the installer for your operating system.
3. Run the installer, `xsetup`, or `xsetup.exe`, which opens the Welcome page. Extract the installer package.
4. Click **Next** to open the Select Install Type page of the Installer.
5. If installing with the web installer, enter your AMD user account credentials, and select **Download and Install Now** (only needed by web installer).
6. Click **Next** to open the Accept License Agreements page of the Installer.
7. Accept the terms and conditions by clicking each **I Agree** check box.
8. Click **Next** to open the Select Product to Install page of the Installer.
9. Select **Vitis** and click **Next** to open the Vitis Unified Software Platform page of the Installer.
10. Customize your installation by selecting design tools and devices (optional).

The default Design Tools selections are for standard Vitis Unified Software Platform installations, and include Vitis, Vivado, and Vitis HLS. You do not need to separately install Vivado tools. You can also install Model Composer and System Generator if needed.

You can enable **Vitis IP Cache** to install cache files for example designs found in the release. This is not required, but when selected, the files are installed at `<installdir>/Vitis/<release>/data/cache/xilinx`.

The default Devices selections are for devices used on standard platforms supported by the Vitis tools. You can disable some devices that might not be of interest in your installation.



IMPORTANT! For the accelerated flow, do not deselect the following option. It is required for installation.

- **Devices → Install devices for Alveo and Edge acceleration platforms**

11. Click **Next** to open the Accept License Agreements page of the Installer and accept as appropriate.
12. Click **Next** to open the Select Destination Directory page of the Installer.
13. Specify the installation directory, review the location summary, review the disk space required to insure there is enough space, and click **Next** to open the Installation Summary page of the Installer.
14. Click **Install** to begin the installation of the software.

After a successful installation of the full Vitis unified software, a confirmation message is displayed, with a prompt to run the `installLibs.sh` script.

1. Locate the script at: `<install_dir>/Vitis/<release>/scripts/installLibs.sh`, where `<install_dir>` is the location of your installation, and `<release>` is the installation version.

Note: This script is not required on Windows for the embedded flow.

2. Run the script using `sudo` privileges as follows:

```
sudo installLibs.sh
```

The command installs a number of necessary packages for the Vitis tools based on the OS of your system.



IMPORTANT! Pay attention to any messages returned by the script. You might need to install any missing packages manually. For example, if your installation of Linux does not include the `zip` command-line utility, you need to manually install it. The utility is required by some of the Vitis tools and the `installLibs.sh` script does not install it for you.

Setting Up the Environment to Run the Vitis Software Platform

Accelerated Software Development

To configure the environment to run the Vitis unified software platform for the embedded flow, run the following scripts in a command shell to set up the tools to run in that shell:

```
#set up XILINX_VITIS and XILINX_VIVADO variables
source <Vitis_install_path>/Vitis/2024.2/settings64.sh
#set up XILINX_XRT
source /opt/xilinx/xrt/setup.sh
```

To configure the environment to run the Vitis unified software platform for the accelerated flow, run the following scripts in a command shell to set up the tools to run in that shell:

```
#set up XILINX_VITIS and XILINX_VIVADO variables
source <Vitis_install_path>/Vitis/2024.2/settings64.sh
```



TIP: *.csh scripts are also provided.*

This sets up the tools for the Vitis application acceleration development flow, the Vitis embedded software development flow, and the AI Engine tools for development on Versal adaptive SoC AI Engine devices.

To use any platforms you have downloaded as described in the [Installing Xilinx Runtime and Platforms](#), set the following environment variable to point to the location of the platforms. The following is more common in the Accelerated Flow.

```
export PLATFORM_REPO_PATHS=<path to platforms>
```

This identifies the location of platform files for the tools, and makes them accessible to your design projects.

Embedded Software Development

To configure the environment to run the Vitis software platform on the embedded flow, run the following script in a command shell to set up the tools to run in that shell:

```
#setup XILINX_VITIS and XILINX_VIVADO variables
source <Vitis_install_path>/Vitis/2024.2/settings64.sh
```



TIP: *.csh scripts are also provided.*

This sets up the tools for the Vitis embedded software development flow.

To run Vitis, type **vitis** and press enter in the Console.

If you have created a shortcut on your desktop, you can also double the shortcut to launch Vitis unified IDE.

Windows

To launch the Vitis software platform from Windows, do one of the following:

- Launch from a desktop button or Start menu command.
- From a Windows command shell, use `settings64.bat`:

```
C:> <VITIS_INSTALL_DIR>\VITIS\2024.2\settings64.bat
```

And launch: `vitis`.

Additional Resources and Legal Notices

Finding Additional Documentation

Technical Information Portal

The AMD Technical Information Portal is an online tool that provides robust search and navigation for documentation using your web browser. To access the Technical Information Portal, go to <https://docs.amd.com>.

Documentation Navigator

Documentation Navigator (DocNav) is an installed tool that provides access to AMD Adaptive Computing documents, videos, and support resources, which you can filter and search to find information. To open DocNav:

- From the AMD Vivado™ IDE, select **Help** → **Documentation and Tutorials**.
- On Windows, click the **Start** button and select **Xilinx Design Tools** → **DocNav**.
- At the Linux command prompt, enter `docnav`.

Note: For more information on DocNav, refer to the *Documentation Navigator User Guide* ([UG968](#)).

Design Hubs

AMD Design Hubs provide links to documentation organized by design tasks and other topics, which you can use to learn key concepts and address frequently asked questions. To access the Design Hubs:

- In DocNav, click the **Design Hubs View** tab.
- Go to the [Design Hubs](#) web page.

Support Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see [Support](#).

References

These documents provide supplemental material useful with this guide:

1. *Data Center Acceleration using Vitis* ([UG1700](#))
 2. *Embedded Design Development Using Vitis* ([UG1701](#))
 3. *Vitis Reference Guide* ([UG1702](#))
 4. *Introduction to FPGA Design with Vivado High-Level Synthesis* ([UG998](#))
 5. *Vitis Unified Software Platform Documentation: Embedded Software Development* ([UG1400](#))
 6. *Vitis High-Level Synthesis User Guide* ([UG1399](#))
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Revision History

The following table shows the revision history for this document.

Section	Revision Summary
11/13/2024 Version 2024.2	
Initial release.	N/A

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